1

We know,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$= 9\times10^9 \times \frac{3.12\times10^{-6}\times1.48\times10^{-6}}{(0.123)^2}$$

$$= 2.75\ N$$

attractive force

(Ans:)

Here,

$q_1 = +3.12\times10^{-6}\ C$

$q_2 = -1.48\times10^{-6}\ C$

$r = 12.3\ cm$

$= 0.123\ m$

$F = ?$

2

We know,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$\Rightarrow r^2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{F}$$

$$= 9\times10^9 \times \frac{26.3\times10^{-6}\times47.1\times10^{-6}}{5.66}$$

$$= 1.97$$

$$\therefore r = 1.40\ m$$

(Ans:)

Here,

$q_1 = 26.3\ \mu C$

$= 26.3\times10^{-6}\ C$

$q_2 = -47.1\ \mu C$

$= -47.1\times10^{-6}\ C$

$F = 5.66\ N$

$r = ?$

3

We know

$F = qE$

$E = \frac{F}{q}$

$= \frac{mg}{q}$

$= \frac{6.64 \times 10^{-27} \times 9.8}{2 \times 1.6 \times 10^{-19}}$

$= 2.03 \times 10^{-7}\ NC^{-1}$

Direction of the electric field is upward.

Here,

Mass of He nucleus

$m = 6.64 \times 10^{-27}$ kg

charge $q = 2e$

$= 2 \times 1.6 \times 10^{-19}$ C

$E = ?$

$g = 9.8\ ms^{-2}$

4

a

We know

$F = q_1 E$

$E = \frac{F}{q_1} = \frac{3.0 \times 10^{-6}}{2 \times 10^{-9}}$

$= 1.5 \times 10^{3}$ N/C

E is upward.

Here,

Particle charge

$q_1 = -2.0 \times 10^{-9}$ C

downward electric force

$F = 3.0 \times 10^{-6}$ N

b

For a proton, charge is $q_2 = 1.6 \times 10^{-19}$ C

Electric force, $F_e = q_2 E$

$= 1.6 \times 10^{-19} \times 1.5 \times 10^{3}$

$= 2.4 \times 10^{-16}$ N

F is attractive ~~force~~ force on the direction of particle.

c

For gravitational force, on the proton,

$F_g = mg$

$= 1.67 \times 10^{-27} \times 9.8$

$= 1.6 \times 10^{-26}\ N$

$m = 1.67 \times 10^{-27}\ kg$

$g = 9.8\ ms^{-2}$

$F = ?$

d

$$\frac{F_e}{F_g} = \frac{2.4 \times 10^{-16}}{1.6 \times 10^{-26}} = 1.5 \times 10^{10}$$

$\boxed{5}$ We know,

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\Rightarrow 2.30 = 9 \times 10^9 \times \frac{q}{(0.75)^2}$$

$$\Rightarrow q = \frac{2.30 \times (0.75)^2}{9 \times 10^9}$$

$$\therefore q = \cancel{1.44}\ 1.44 \times 10^{-10}\ C$$

Here,

$r = 75\ cm$

$= 0.75\ m$

$E = 2.30\ N/c$

$q = ?$

[6] The dipole moment

$p = 2aq$

$= 4.30 \times 10^{-9} \times 1.6 \times 10^{-19}$

$= 6.88 \times 10^{-28}$ C.m

Here,

$q = 1.6 \times 10^{-19}$ C

$2a = 4.30$ nm

$= 4.30 \times 10^{-9}$ m

$p = ?$

[7]

We know,

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{x^3}$$

$$= 9 \times 10^{9} \times \frac{3.56 \times 10^{-29} \text{ C.m}}{(25.4 \times 10^{-9})^3}$$

$= 1.95 \times 10^{4}$ N/C

Here,

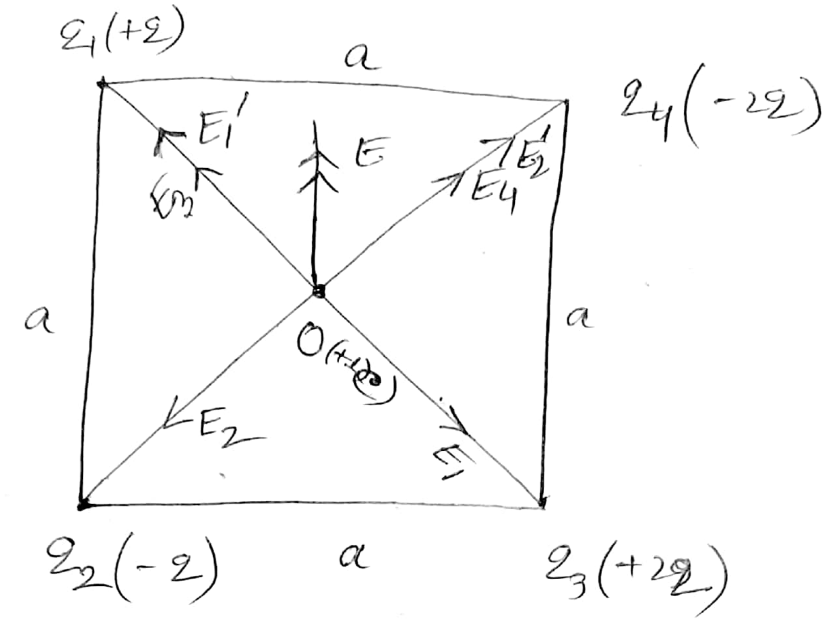
$p = 3.56 \times 10^{-29}$ C.m

$x = 25.4$ nm

$= 25.4 \times 10^{-9}$ m

$E = ?$

$\boxed{8}$



Here,

$a = 5.20\ cm = 0.052\ m$

$q = 11.8\ nc = 11.8 \times 10^{-9} c$

Let us consider, r ~~is~~ is the distance between charge and centre of the square.

$$\therefore r = \sqrt{a^2+a^2}/2$$

$$= \frac{\sqrt{2a^2}}{2}$$

$$= \frac{\sqrt{2}a}{2}$$

$$= \frac{a}{\sqrt{2}}.$$

For charge q_1, Electric field $E_1 = \frac{1}{4\pi\epsilon_0}\frac{q_1}{r^2}$

For charge q_2, Electric field $E_2 = \frac{1}{4\pi\epsilon_0}\frac{q_2}{r^2}$

For, q_3, $E_3 = \frac{1}{4\pi\epsilon_0}\frac{q_3}{r^2}$

For, q_4, $E_4 = \frac{1}{4\pi\epsilon_0}\frac{q_4}{r^2}$

Now,
The resultant electric field intensity between E_1 and E_3 is

$$E_1' \cancel{E_1'} = \sqrt{E_1^2 + E_3^2 + 2E_1E_3\cos 180^\circ}$$

$$= \sqrt{E_1^2 + E_3^2 - 2E_1E_3}$$

$$= \sqrt{(E_3 - E_1)^2} \quad [\because E_3 > E_1]$$

$$= E_3 - E_1$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q_3}{r^2} - \frac{1}{4\pi\epsilon_0}\frac{q_1}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}(q_3 - q_1)$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}(2q - q)$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$$

Again,
The resultant electric field intensity between E_2 and E_4 is

$$E_2' = \sqrt{E_2^2 + E_4^2 + 2E_2E_4\cos 180^\circ}$$

$$= E_4 - E_2$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}[q_4 - q_2] \quad \begin{bmatrix} |q_2| = q \\ |q_4| = 2q \end{bmatrix}$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}[2q - q]$$

$$= \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$$

The resultant electric field between E_1' and E_2' is

$$E = \sqrt{E_1'^2 + E_2'^2 + 2E_1' E_2' \cos 90^\circ}$$

$$= \sqrt{E_1'^2 + E_2'^2} \quad [\because E_1' = E_2']$$

$$= \sqrt{2E_1'^2}$$

$$= \sqrt{2}\, E_1'$$

$$= \sqrt{2} \cdot \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$= \sqrt{2} \times \frac{1}{4\pi\epsilon_0} \times \frac{q}{\left(\frac{a}{\sqrt{2}}\right)^2}$$

$$= \sqrt{2} \times 9 \times 10^9 \times \frac{2 \times q}{a^2}$$

$$= \sqrt{2} \times 9 \times 10^9 \times \frac{2 \times 11.8 \times 10^{-9}}{(0.052)^2}$$

$$= 1.11 \times 10^5 \text{ N/C}$$

(Ans.)

$\boxed{9}$

We know,

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$q = \frac{Vr}{9\times10^9}$$

$$= \frac{100\times0.1}{9\times10^9}$$

$$= 1.11\times10^{-9}\ c$$

Ans:

Here,

$r = 10$ cm

$= 0.1$ m

$V = 100$ Volt

$q = ?$

$\boxed{10}$

We know,

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$= 9\times10^9\times\frac{79\times1.6\times10^{-19}}{6.6\times10^{-15}}$$

$$= 1.7\times10^7 \text{ volts.}$$

(Ans:)

Here,

$r = 6.6\times10^{-15}$ m

Atomic number of gold is $Z = 79$

$V = ?$

Scanned with CamScanner

11

We know,

Potential energy

$$U = \frac{1}{4\pi\epsilon_0}\frac{q^2}{r}$$

$$= 9\times10^{9}\times\frac{(1.6\times10^{-19})^2}{6.0\times10^{-15}}$$

$$= 3.84\times10^{-14}\ J$$

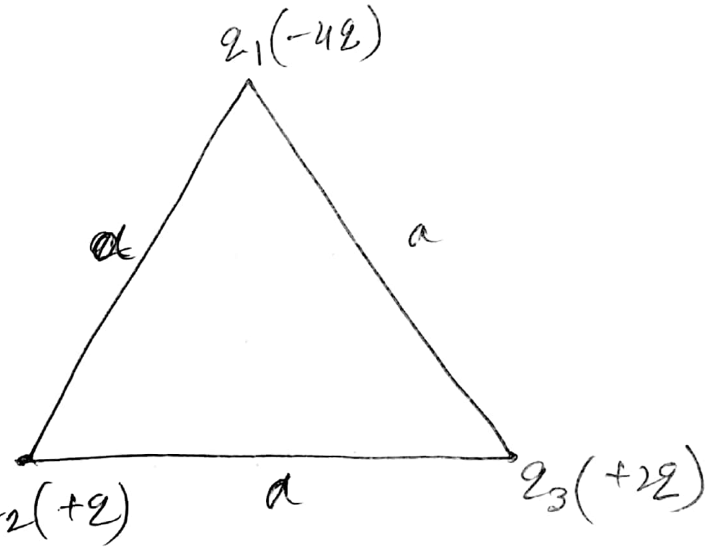
Here,

$q = 1.6\times10^{-19}\ c$

$r = 6.0\times10^{-15}\ m$

$U = ?$

12



$$U = U_{12} + U_{23} + U_{31}$$

$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{(-4q)(q)}{a} + \frac{1}{4\pi\epsilon_0}\frac{(+q)(+2q)}{a} + \frac{1}{4\pi\epsilon_0}\frac{(+2q)(-4q)}{a}$$

$$\Rightarrow U = \frac{1}{4\pi\epsilon_0}\cdot\frac{1}{a}\left(-4q^2 + 2q^2 - 8q^2\right)$$

$$= \cancel{9\times10^{9}\times\frac{1}{1.0\times10^{-7}}\left(-4(\ \ -10q^2\right)}$$

$$= \cancel{9\times10^{9}\times\frac{1}{1.0\times10^{-7}}\times(-10(}$$

Here,

$a = 10\ cm$
$= 0.1\ m$

$q = 1.0\times10^{-7}\ c$

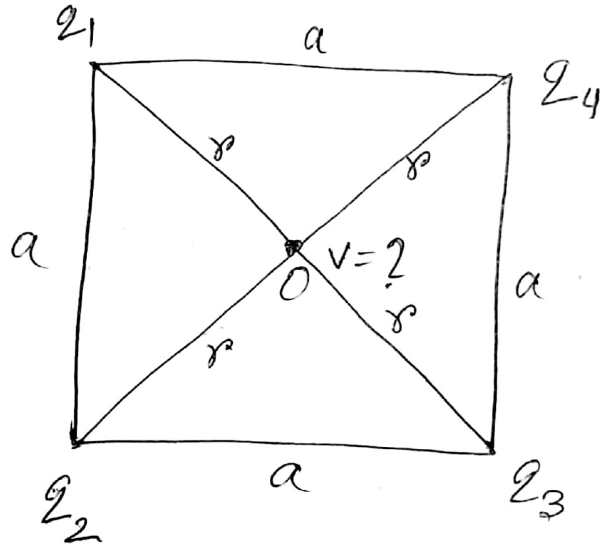
$$U = \frac{1}{4\pi\epsilon_0} \frac{(-10q^2)}{a}$$

$$= 9\times10^{9} \times \frac{(-10)(1.0\times10^{-7})^2}{0.10}$$

$$= -9\times10^{-3}\ J.$$

(Ans:))

$\boxed{13}$



Here,

$q_1 = +1.0\times10^{-8}\ c$

$q_2 = -2.0\times10^{-8}\ c$

$q_3 = +3.0\times10^{-8}\ c$

$q_4 = +2.0\times10^{-8}\ c$

and $a = 1\ m$.

Let the distance r between centre and charge each corner of square.

$$r = \frac{\sqrt{a^2+a^2}}{2} = \frac{\sqrt{2a^2}}{2} = \frac{a\sqrt{2}}{\sqrt{2}\sqrt{2}} = \frac{a}{\sqrt{2}}$$

At the centre potential, $V = V_1 + V_2 + V_3 + V_4$

$$= \frac{1}{4\pi\epsilon_0}\frac{q_1}{r} + \frac{1}{4\pi\epsilon_0}\frac{q_2}{r} + \frac{1}{4\pi\epsilon_0}\frac{q_3}{r} + \frac{1}{4\pi\epsilon_0}\frac{q_4}{r}$$

Scanned with CamScanner

$$V = \frac{1}{4\pi\epsilon_0}\frac{1}{r}(q_1+q_2+q_3+q_4)$$

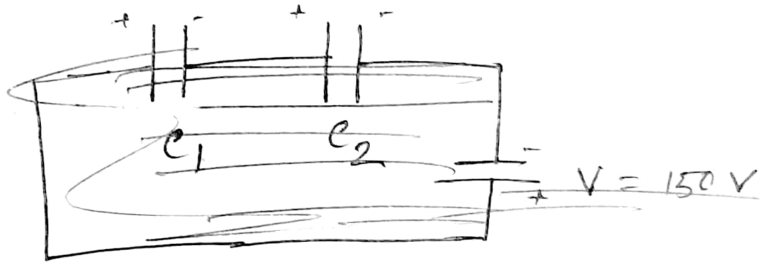
$$= 9\times10^{9}\times\frac{1}{\frac{a}{\sqrt{2}}}(1-2+3+2)\times10^{-8}$$

$$= 9\times10^{9}\times\frac{\sqrt{2}}{1}\times4\times10^{-8}$$

$$= 509.11 \text{ volt.}$$

Ans:

$\boxed{14.}$



Here, $C_1 = \frac{q}{V}$

$$= \frac{50\times10^{-6}}{150}$$

$$= \frac{1}{3}\times10^{-6}\text{ F}$$

Here, $q = 50\ \mu C$

$= 50\times10^{-6}$ C

$V = 150$ V

$$C_2 = 4C_1 = 4\times\frac{1}{3}\times10^{-6}\text{ F}$$

$$= \frac{4}{3}\times10^{-6}\text{ F}$$

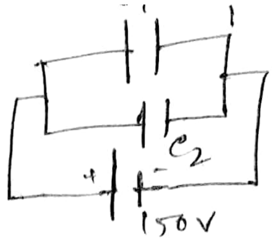
Before joining.

Energy $U_1 = \frac{1}{2}C_1V^2$

$$= \frac{1}{2}\times\frac{1}{3}\times10^{-6}\times(150)^2$$

$$= 37.5\times10^{-4}\text{ J.}$$

After joining



Equivalent capacitance,

$$C_p = C_1 + C_2$$
$$= \left(\frac{1}{3} + \frac{4}{3}\right) \times 10^{-6}\ F$$
$$= \frac{5}{3} \times 10^{-6}\ F$$

$$\cancel{\text{Energy}, U = \frac{1}{2} CV^2}$$
$$\cancel{= \frac{1}{2} \times \frac{5}{3} \times 10^{-6} \times (150)^2}$$

Voltage, $V = \dfrac{\text{total charge}}{\text{total capacitance}}$

$$= \frac{50 \times 10^{-6}}{\frac{5}{3} \times 10^{-6}}$$

$$= 30\ V$$

Energy, $U = \frac{1}{2} CV^2$

$$= \frac{1}{2} \times \frac{5}{3} \times 10^{-6} \times (30)^2$$
$$= 7.5 \times 10^{-4}\ J$$

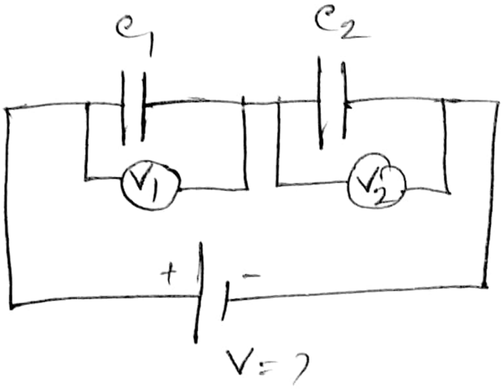
Loss of energy $= (37.5 - 7.5) \times 10^{-4}\ J$

$$= 30 \times 10^{-4}\ J \quad \text{(Ans:)}$$

Scanned with CamScanner

15.

The series combination is shown in figure.



Since charge across each capacitor is the same,

So, $C_1 = \frac{q}{V_1}$ and $C_2 = \frac{q}{V_2}$

$\Rightarrow q = C_1V_1$ $\quad q = C_2V_2$

$\therefore C_1V_1 = C_2V_2$

$\Rightarrow \frac{V_2}{V_1} = \frac{C_1}{C_2}$ ....(1)

Now,
Air capacitor, $C_1 = \frac{\varepsilon_0 A}{d} = \varepsilon_0 \times \frac{100\times10^{-4}}{4\times10^{-3}} = 2.5\,\varepsilon_0$

$C_2 = \frac{\varepsilon_0 \varepsilon_r A}{d} = \frac{\varepsilon_0 \times 2.5\times20\times10^{-4}}{0.1\times10^{-3}}$

$= 50\,\varepsilon_0$

For C_2 capacitor,
$\varepsilon_r = 2.5$
$A = 20\ cm^2$
$d = 0.1\ mm$

Now, we get from eqn (1)

$\frac{V_2}{V_1} = \frac{2.5\,\varepsilon_0}{50\,\varepsilon_0} = 0.05$

$\therefore V_2 = 0.05\,V_1$

$= 0.05 \times 100$

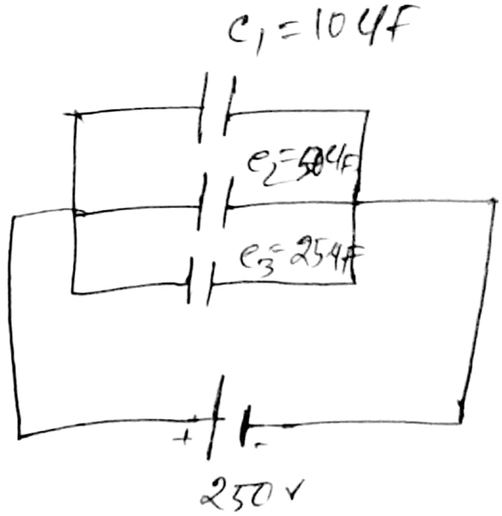
$= 5\,V$

For air capacitor
$V_1 =$ ~~50V~~ 100 V

Hence, $V = V_1 + V_2$
$= 100 + 5$
$= 105\ V$ Ans.

16

(i) Parallel connection

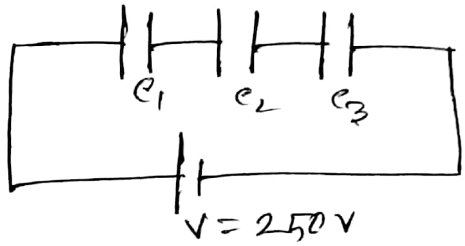


$Q_1 = C_1 V$
$= 10 \times 250$
$= 2500\ \mu C$

$Q_2 = C_2 V$
$= 50 \times 250$
$= 12500\ \mu C$

$Q_3 = C_3 V = 25 \times 250 = \cancel{6750}\ 6250\ \mu C$

(ii) In ~~series~~ Parallel connection,



$C_p = C_1 + C_2 + C_3$
$= 10 + 50 + 25$
$= 85\ \mu F$

In ~~parallel~~ series connection

$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$
$= \frac{1}{10} + \frac{1}{50} + \frac{1}{25}$
$\therefore C_p = 6.25\ \mu F$

(iii)

$\cancel{V_T} = Q = C_s V$

$= 6.25 \times 250$

$= 1562.5\ \mu C$

$$\left.\begin{aligned} V_1 &= \frac{Q}{C_1} = \frac{1562.5}{10} = 156.25\ V \\ V_2 &= \frac{Q}{C_2} = \frac{1562.5}{25} = 62.5\ V \\ V_3 &= \frac{Q}{C_3} = \frac{1562.5}{50} = 31.25\ V \end{aligned}\right\} \text{Ans:}$$

$\boxed{17}$ Here, $C_1 = 4\ \mu F$

$C_2 = 2\ \mu F$

When joined in series,

$C_1 V_1 = C_2 V_2$

$\Rightarrow 4V_1 = 2V_2$

$\Rightarrow V_2 = 2V_1$

$V_1 + V_2 = 100$

$\Rightarrow V_1 + 2V_1 = 100$

$\Rightarrow 3V_1 = 100$

$V_1 = \frac{100}{3} = 33.33\ V$

$V_2 = 2 \times 33.33$

$= 66.66\ V$

$Q_1 = Q_2 = \left(\frac{200}{3}\right) \times 2 = \frac{400}{3}\ \mu C$

$\therefore$ ~~Total~~

Scanned with CamScanner

When joined in parallel

Total charge $Q = \frac{400}{3}\ \mu C$

Total capacitance $C = 4+2 = 6\ \mu F$

$$C = \frac{Q}{V}$$

$$V = \frac{Q}{C} = \frac{\frac{400}{3}}{6}$$

$$= \frac{400}{3 \times 6} = \frac{400}{18}$$

$$Q_1 = C_1 V$$

$$= 4 \times \frac{400}{18} = 88.89\ \mu C$$

$$Q_2 = C_2 V = 2 \times \frac{400}{18} = \frac{800}{18} = 44.44\ \mu C$$

$\boxed{18}$

Here,

$d = 1mm = 1 \times 10^{-3} m$

$C = 1F$

$A = ?$

We know

$$C = \frac{\epsilon_0 A}{d}$$

$$A = \frac{Cd}{\epsilon_0}$$

$$= \frac{1 \times 10^{-3}}{8.854 \times 10^{-12}}$$

$$= 1.12 \times 10^{8}\ m^2$$

(Ans:)

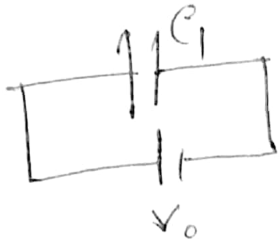
Scanned with CamScanner

$\boxed{10}$

(a)

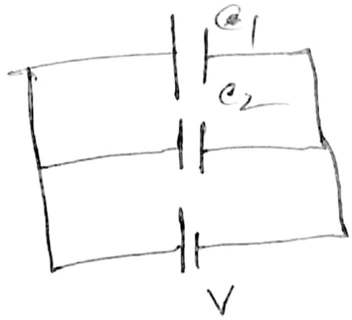
$Q_0 = CV_0$

In Initial



$Q_1 = C_1V$

$Q_2 = C_2V$



$\therefore Q_0 = Q_1 + Q_2$

$\Rightarrow C_1V_0 = C_1V + C_2V$

$\Rightarrow V(C_1 + C_2) = C_1V_0$

$$\therefore V = \frac{C_1V_0}{C_1 + C_2}$$

$\boxed{b}$ Before joining

Energy $U_0 = \frac{1}{2} C_1 V_0^2$

After joining

Energy $U = \frac{1}{2} C_p V^2$ $\left[\cancel{C_p = C_1+} \quad C_p = C_1 + C_2\right]$

$$\Rightarrow U = \frac{1}{2}(C_1 + C_2)\left(\frac{C_1V_0}{C_1 + C_2}\right)^2$$

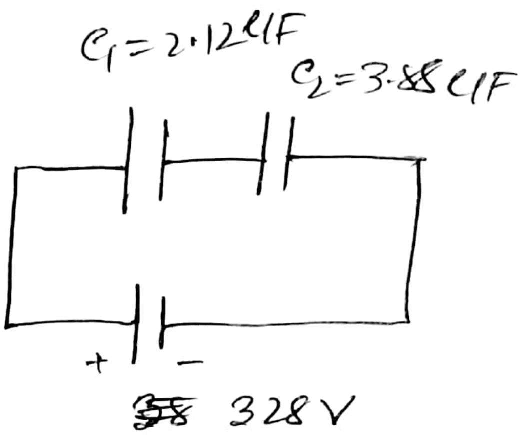
$$= \frac{1}{2} C_1^2 V_0^2 \frac{1}{C_1 + C_2}$$

$$U = \frac{1}{2} C_1 V_0^2 \left(\frac{C_1}{C_1 + C_2}\right)$$

$$= V_0 \left(\frac{C_1}{C_1 + C_2}\right)$$

Ans.

$\boxed{20}$



$$C_s = \left(\frac{1}{C_1} + \frac{1}{C_2}\right)^{-1}$$

$$= \left(\frac{1}{2.12} + \frac{1}{3.88}\right)^{-1}$$

$$= 1.37\,\mu F$$

$$= 1.37 \times 10^{-6}\,F$$

Here
$C_1 = 2.12\,\mu F$
$C_2 = 3.88\,\mu F$
$V = 328\,V$

The total energy stored in capacitor

$$U = \frac{1}{2} C V^2$$

$$= \frac{1}{2} \times 1.37 \times 10^{-6} \times (328)^2$$

$$= 7.4 \times 10^{-2} \text{ joule}.$$